must be conserved in the system, and thus

$$p_1 + p_2 = p_3 + p_4 \tag{5.6}$$

The connection between kinetic energy E_{kin} and linear momentum is given by the relation

$$E_{\text{kin}} = p^2/(2m) \tag{5.7}$$

(c) The *total charge* (protons and electrons) of the system must be constant, i.e.

$$Z_1 + Z_2 = Z_3 + Z_4 \tag{5.8}$$

where the charge is in electron units.

(d) The *mass number* (number of nucleons) in the system must be constant, i.e.

$$A_1 + A_2 = A_3 + A_4 \tag{5.9}$$

(e) The *total angular momentum* p_{I} of the system must be conserved, i.e.

$$(p_{\text{I}})_1 + (p_{\text{I}})_2 = (p_{\text{I}})_3 + (p_{\text{I}})_4 \tag{5.10}$$

Since there exist two types of angular momentum, one caused by orbital movement of the individual nucleons and the other due to the intrinsic spin of the nucleons (internal angular momentum), a more practical formulation of (5.10) is

$$\Delta I = I_3 + I_4 - I_1 - I_2 \tag{5.11}$$

where I is the (total) *nuclear spin quantum number.* The quantum rule is

$$\Delta I = 0,\ \pm 1,\ \pm 2,\ \pm 3, \ldots \tag{5.12}$$

i.e. the change of nuclear spin in a reaction must have an integral value.

The three first laws are general in classical physics; the last two refer particularly to nuclear reactions. In Ch. 6 and 11 other conservation laws are discussed for nuclear reactions, but these are less important in radioactive decay.

5.3. ALPHA DECAY

5.3.1. Detection

Alpha particles cause extensive ionization in matter, as discussed in more detail in Ch 8. If the particles are allowed to pass into a gas, the electrons released by the ionization can be collected on a positive electrode to produce a pulse or current. *Ionization chambers* and *proportional counters* are instruments of this kind, which permit the individual counting of each α-particle emitted by a sample. Alpha particles interacting with matter may also cause molecular excitation, which can result in fluorescence. This fluorescence – or

scintillation – allowed the first observation of individual nuclear particles. The ionization in semiconductors caused by α-particles is now the most common means of detection, see §9.4.

5.3.2. Decay energy

Alpha decay is observed for the elements heavier than lead and for a few nuclei as light as the lanthanide elements. However, ^{8}Be can also undergo alpha decay, i.e. daughter and alpha particle are equal. Alpha decay can be written symbolically as

$$ {}^{A}_{Z}\mathrm{X} \;\rightarrow\; {}^{A-4}_{Z-2}\mathrm{X} + {}^{4}_{2}\mathrm{He} \tag{5.13} $$

We use X to indicate any element defined by its nuclear charge, Z and Z-2 in this equation. Examples are given in Ch. 1, and can be found e.g. in the natural radioactive decay series, see §13.5.2.

The decay energy can be calculated from the known atomic masses, because the binding energy released (spontaneous decay processes must be exoergic) corresponds to a disappearance of mass, cf. eqns. (4.5) and (4.10). This energy is also called the *Q-value of the reaction*

$$ Q\,(\mathrm{MeV}) = -931.5\;\Delta M\,(\mathrm{u}) \tag{5.14} $$

For α-decay we can define the Q-value as

$$ Q_\alpha = -931.5\left(M^{A-4}_{Z-2} + M_{\mathrm{He}} - M^{A}_{Z}\right) \tag{5.15} $$

We always write the products minus the reactants within the parenthesis. A decrease in total mass in α-decay means a release of energy. The minus sign before the constant 931.5 MeV/u, cf Ch. 3, is necessary to make Q positive for spontaneous decay.

Consider the following example. For the decay reaction $^{238}\mathrm{U} \rightarrow {}^{234}\mathrm{Th} + {}^{4}\mathrm{He}$, the mass values for ^{238}U and ^{4}He are found in Table 3.1; for ^{234}Th it is 234.043 594 u. Thus we obtain $Q_\alpha = -931.5 \times (234.043\,594 + 4.002\,603 - 238.050\,7785) = 4.274$ MeV.

If the products are formed in their ground states, which is common for α-decay, the total decay energy is partitioned into the kinetic energies of the daughter nucleus (E^{A-4}_{Z-2}) and the helium nucleus (E_α):

$$ Q_\alpha = E^{A-4}_{Z-2} + E_\alpha \tag{5.16} $$

Because of conservation of energy (5.4) and momentum (5.6)

$$ E_{Z-2} = Q_\alpha M_\alpha / M^{A}_{Z} \tag{5.17} $$

and

$$ E_\alpha = Q_\alpha\, M^{A-4}_{Z-2} / M^{A}_{Z} \tag{5.18} $$

From these equations we can calculate the kinetic energy of the ^{234}Th daughter to be 0.072 MeV, while that of the α-particle is 4.202 MeV. Because of the large mass difference between the α-emitting nucleus and the helium atom, almost all of the energy is carried away with the α-particle.

Although the kinetic energy of the daughter nucleus is small in comparison with that of the α-particle, it is large (72 000 eV) in comparison with chemical binding energies (<5 eV). Thus the recoiling daughter easily breaks all chemical bonds by which it is bound to other atoms.

In 1904 it was observed by H. Brooks that measurements on ^{218}Po (RaA), obtained from radon, led to a contamination of the detection chamber by ^{214}Pb (RaB) and ^{214}Bi (RaC). This was explained by Rutherford as being due to daughter recoil in the α-decay of ^{218}Po in the sequence (written symbolically):

$$^{222}\mathrm{Rn}\left(\alpha, 3.8\ \mathrm{d}\right)\,^{218}\mathrm{Po}\left(\alpha, 3.05\ \mathrm{min}\right)\,^{214}\mathrm{Pb}\left(\beta^{-}, 27\ \mathrm{min}\right)\,^{214}\mathrm{Bi}\left(\beta^{-}, 20\ \mathrm{min}\right)\ldots \quad (5.19)$$

This recoil led to ejection of ^{214}Pb into the wall of the instrument. The use of the *recoil* of the daughter to effect its separation was employed by O. Hahn beginning in 1909 and played a central role in elucidating the different natural radioactive decay chains.

The recoil may affect such chemical properties as the solubility or dissolution rate of compounds. For example the dissolution of uranium from uranium rich minerals is considerably higher than one would expect from laboratory solubility data because α and U-atom recoil have moved U-atoms away from their normal sites in the mineral.

Alpha-decay energies are most precisely measured in magnetic spectrometers. From (3.6) and (3.11) it is calculated that

$$E_\alpha = 2\mathbf{e}^2 B^2 r^2 / m_{\mathrm{He}} \quad (5.20)$$

From knowledge of the values of **e**, m_{He}, B, and r, E_α can be calculated. A more common technique is to use *semiconductor detectors* combined with pulse height analyzers ("α-spectrometers", Ch. 9).

5.4. BETA DECAY

5.4.1. Detection

Energetic electrons cause ionization and molecular excitation in matter, although the effect is weaker and more difficult to detect than for α-particles. As a result the effect must be amplified for counting individual β-particles. Ionization is used in *proportional* and *Geiger counters*. Scintillation counting can also be used with various detector systems (§9.5).